



Stock returns revisited and variances hedged by machine learning

Dilip B. Madan & King Wang

To cite this article: Dilip B. Madan & King Wang (2026) Stock returns revisited and variances hedged by machine learning, Quantitative Finance, 26:3, 341-354, DOI: [10.1080/14697688.2025.2609651](https://doi.org/10.1080/14697688.2025.2609651)

To link to this article: <https://doi.org/10.1080/14697688.2025.2609651>



Published online: 02 Feb 2026.



Submit your article to this journal [↗](#)



Article views: 150



View related articles [↗](#)



View Crossmark data [↗](#)

Stock returns revisited and variances hedged by machine learning

DILIP B. MADAN^{*†} and KING WANG[‡]

[†]Robert H. Smith School of Business, University of Maryland, College Park, MD 20742, USA

[‡]Derivative Product Strats, Morgan Stanley, 1585 Broadway 2nd Floor, New York, NY 10036, USA

(Received 14 February 2025; accepted 19 December 2025)

The empirically supported property of absolute variations dominating quadratic variations are employed to motivate the construction of models with percentage returns bounded by unity in absolute value. Characteristic functions are developed for the log price relative for the new return models. The models are estimated on time series and option data and demonstrate improvements delivered by the inverse logistic transformation. Applications to pricing return variations and options on them are developed. Hedging strategies use Machine Learning methods on simulated sample spaces. It is observed that the log contract hedge introduces a high volatility dynamic hedge that theoretically may be compensated by the log contract leaving the variance contract as a residual. The Machine Learned hedge works with other functions estimated here by a Gaussian Process Regression that substantially reduces the volatility of the dynamic hedge and yet leaves, approximately, the variance payout as the residual.

Keywords: Bilateral gamma model; The CGMY model; Gaussian process regression; Custom kernel

JEL Classification: G10, G11, G12

1. Introduction

Bachelier (1900) modeled stock returns as a random process in continuous time by Brownian motion. Brownian motion is a process with independent and identically distributed increments with continuous sample paths and normally distributed increments over regular time steps. The normal distribution permits stock prices to go negative with positive probability. With a view to incorporating the limited liability status of stock prices, Samuelson (1965) applied the Brownian motion model to the logarithm of the stock price. Black and Scholes (1973) and Merton (1973) went on to develop the theory of option pricing for such a process. Bachelier (1914) already observed the absence of normality in stock returns over a horizon. The evidence of nonnormality has led researchers to develop other models of independent and identical increments or Lévy processes for the stock price motion. Mandelbrot (1963) proposed the process of the stable distribution for the increments. Other models include the variance gamma model of Madan and Seneta (1990) and Madan *et al.* (1998), the normal inverse Gaussian model of Barndorff-Nielsen (1998), the generalized hyperbolic of Eberlein (2001),

the CGMY model of Carr *et al.* (2002) and the bilateral gamma model of Küchler and Tappe (2008). Excluding, the stable distribution that lacks exponential moments, these models have been applied to the problem of option pricing with calibrations to market data facilitated by Fourier inversion techniques (Carr and Madan 1999). In all cases, the models apply to returns defined as the log price relative over two time points.

The starting point of this paper is the observation that empirically return quadratic variations are dominated by their variations but the logarithmic return models permit violations of this property. The property of dominating quadratic variations by variations is satisfied when returns are below unity in absolute value. Just as the normal distribution permitted negative stock prices with positive probability real-valued logarithmic returns x , permit returns $(e^x - 1)$ to exceed unity with positive probability for $x > \ln(2)$. For 586 stocks over 3838 days from 3 January 2007 to 3 December 2022 for a total of 2,249,068 stock days the absolute return exceeded unity on 0 days. For horizons shorter than a day, and especially if processes approximate in any realistic way a continuous process, one anticipates zero violations for absolute returns exceeding unity. Discontinuous approximations to continuity focus attention on modeling a large number of small jumps.

*Corresponding author. Email: dbm@umd.edu, dbm@rhsmith.umd.edu

Such observations motivate the consideration of models with absolute returns bounded by unity. From this perspective, the use of logarithmic returns may be seen as essentially motivated by analytical tractability. The models developed here for alternative return specifications require numerical approaches in their implementation but these are now well within current computational abilities. We go on to build and report on the performance delivered by these alternatives.

On daily market close price data, inverse logistic and inverse hyperbolic tangent returns are entertained as alternatives to exponential or inverse logarithmic returns. It is observed that inverse logistic returns deliver a superior performance to inverse hyperbolic tangent returns while both improve upon the classical exponential model. With regard to option pricing, characteristic functions are formulated in the inverse logistic case and calibrations to *SPY* options deliver an improvement over exponential returns. Bounding returns by a factor below unity is taken up next and a bound of 0.5 delivers an improvement on time series data. For *SPY* option data, the bound of 0.5 is inferior to a bound of unity.

Option pricing is reported on in the inverse logistic case for two classes of Lévy processes. They are the bilateral gamma, and the *CGMY* model with separate *C* and *Y* parameters on the two sides. For the daily return time series data just the bilateral gamma model is reported on.

More recently, the logarithmic return has played a critical role in the development of the market for volatility trading. Strategies for pricing by replication the payout of the realized quadratic variation of the logarithm of the stock price over a term, under assumptions of sample path continuity, were observed in Neuberger (1994). The contract paying the logarithmic return over the term coupled with dynamically trading the stock over the term constituted the replication hedge. The variance swap contract followed and trading began, for example by Goldman Sachs, Merrill Lynch and Morgan Stanley, in the early 1990s. The market for variance swaps has grown substantially with a notional outstanding vega now at 2 billion dollars. The role of the log contract in hedging and pricing the risk has received considerable attention in the literature from a variety of perspectives and we cite by way of examples, Broadie and Jain (2008), Carr and Wu (2006, 2009), Carr and Lee (2010), Carr *et al.* (2012), Carr and Madan (1998), Demeterfi *et al.* (1999), Gatheral (2006), and Davis *et al.* (2013).

Here we consider instead the variance of the simple return or the squared percentage change in place of the squared log price differential. The log contract is then no longer the appropriate hedge function and neither is the dynamic hedge the position of investing two dollars in the stock return. For answers to both these questions, we turn to Machine Learning (*ML*) technologies applied to sample spaces generated by inverse logistic returns driven by a Lévy process. The specific *ML* technology employed is the Gaussian Process Regression (*GPR*) with custom kernels for the implementation of the dynamic stock hedge. The resulting hedge functions differ markedly from the log contract and lead us to develop *GPR* hedges for the traditional variance contract. The *GPR* hedges build other locally quadratic functions with lower volatilities in the associated dynamic hedges permitting lower residual risks compared to the log contract in realistic contexts

admitting the presence of residual risk. The log contract in contrast relies on a theorem, valid in nonexistent circumstances, advocating dynamic hedges with much greater risk, in the hope that the log contract position will compensate them and leave us with only the variance swap payout. The paper thereby offers an advance by delivering competitive *ML*-based hedging technologies.

The outline of the rest of the paper is as follows. Section 2 introduces the inverse logistic and inverse hyperbolic tangent return models. Characteristic functions are developed for these models in Section 3. Results on time series data are presented in Section 4. Section 5 takes up results on option pricing. Section 6 goes on to develop inverse Laplace transform methods for pricing options on realized absolute and variance returns. Machine Learning technologies are developed for hedging in Section 7. Section 8 provides a comparison between simulated variation option prices with Laplace inversion. Section 9 reports on *ML* hedging of variation contracts and their contrast with the log contract hedge. Section 10 concludes.

2. Returns revised

Let $S(t)$ be the path of a stock price in continuous time for $0 < t < T$. We suppose that $S(t)$ is a positive process with an instantaneous mean rate of return of $\eta(t) dt$. The process is defined on a filtered probability space $(\Omega, \{\mathcal{F}_t, 0 \leq t \leq T\}, P)$. The mean deflated price process $M(t)$ is a P martingale with unit initial value, where

$$M(t) = \exp \left(\int_0^t -\eta(u) du \right) \frac{S(t)}{S(0)}. \quad (1)$$

We further suppose that $M(t)$ is a pure jump process with no drift or continuous martingale component. The process $M(t)$ is the stochastic exponential of a compensated jump martingale $m(t)$. Classically for an integer valued random measure $\mu(dx, dt)$ associated with the jumps of the process $X(t)$ with compensator $\nu(dx, dt)$ let

$$m(t) = (e^x - 1) * (\mu - \nu). \quad (2)$$

It follows that

$$M(t) = \exp \left(X(t) - \int_0^t \int_{-\infty}^{\infty} (e^x - 1) \nu(dx, du) \right) \quad (3)$$

and

$$\eta(t) dt = \int_{-\infty}^{\infty} (e^x - 1) \nu(dx, dt). \quad (4)$$

The variation $V(t)$ and quadratic variation $Q(t)$ of returns may be defined by

$$V(t) = \int_0^t \int_{-\infty}^{\infty} |e^x - 1| \mu(dx, ds) \quad (5)$$

$$Q(t) = \int_0^t \int_{-\infty}^{\infty} (e^x - 1)^2 \mu(dx, ds) \quad (6)$$

and we suppose both are finite.

The introduction reported that on a large (586) set stocks over nearly four thousand (3838) days the squared return was dominated by the absolute return consistent with the absolute return being below unity. We anticipate that quite generally for processes that are discontinuous but approximating continuity the quadratic variation should be dominated by the variation. The exponential return of $e^x - 1$ violates this property of $x > \ln(2)$. Most models for $X(t)$ in the literature modeling logarithmic returns have positive arrival rates for jumps $x > \ln(2)$. We consider to continue using the wide class of real valued Lévy processes as models for $X(t)$ but wish to alter the relationship between x and the stock return y that classically is inverse logarithmic or exponential. The exponential correctly bounds the down side by negative unity but is unrealistically unbounded on the up side.

Consider first a logistic relation between x and y whereby

$$y(x) = \frac{2}{1 + e^{-x}} - 1. \quad (7)$$

The return x is then related to the stock return y by

$$x = -\ln \left(\frac{2}{1 + y} - 1 \right). \quad (8)$$

An alternative model is to consider

$$y(x) = \tanh(x) \quad (9)$$

$$x = \frac{1}{2} \ln \left(\frac{1 + y}{1 - y} \right). \quad (10)$$

In either case, we consider the model for $m(t)$ as

$$m(t) = (y(x) * (\mu - \nu))_t \quad (11)$$

Define

$$Y(t) = (y(x) * \mu)_t \quad (12)$$

$$= \int_0^t \int_{-\infty}^{\infty} y(x) \mu(dx, dt) \quad (13)$$

PROPOSITION 1 The stock price is given by

$$S(t) = S(0) \prod_{s \leq t} (1 + \Delta Y(s)) \quad (14)$$

Proof The stochastic exponential of $m(t)$ as defined by equation (11) is given by

$$\mathcal{E}(m(t)) = \exp(-(y(x) * \nu)_t) \prod_{s \leq t} (1 + \Delta Y(s)) \quad (15)$$

$$= \exp \left(- \int_0^t \eta(s) ds \right) \frac{S(t)}{S(0)} \quad (16)$$

where the second equation follows from equation (1) on recognizing that $M(t)$ is the stochastic exponential of $m(t)$.

It follows that

$$\eta(t) dt = \int_{-\infty}^{\infty} y(x) \nu(dx, dt)$$

and $S(t)$ is as per equation (14) ■

3. Characteristic functions

By equation (14), the log price relative is given by

$$\ln \left(\frac{S(t)}{S(0)} \right) = \sum_{s \leq t} \ln(1 + \Delta Y(s)). \quad (17)$$

Suppose now that $X(t)$ is a pure jump Lévy process with compensator

$$\nu(dx, dt) = k(x) dx dt. \quad (18)$$

Let $W(t) = \ln(S(t))$, by equation (17) $W(t)$ is also a Lévy process related to $X(t)$ via

$$\Delta W = \ln(1 + y(\Delta X)). \quad (19)$$

Define

$$w(x) = \ln(1 + y(x)). \quad (20)$$

The support for the Lévy measure for $W(t)$ is $-\infty < w < \ln(2)$ and the jumps in the log price are bounded above by $\ln(2)$. The Lévy density may be derived from $k(x)$ the Lévy density for X by a change of variable formula using the inverse of the function w of equation (20). The simple return now is $(e^w - 1)$ and as it bounded above by unity, w is bounded above $\ln(2)$. The structure is a consequence of enforcing the domination of simple quadratic variation by simple return variation. The logarithm of the stock cannot have an unbounded upside. When $y(x)$ is bounded further to be between ± 0.5 the support of the Lévy density on W is $-\ln(0.5) < w < \ln(1.5)$. The process $X(t)$ has a Lévy density with unbounded support and it is not any longer the logarithm of the stock. Bounds on the simple return naturally bound the support for the logarithm of the stock price. The class of Lévy processes with unbounded supports for the Lévy densities developed in the literature continue to be useful to provide models for the real valued inverse of the simple return. Margin loans for long-short positions delivering a leverage ratio of two, support bounds on the simple return of 0.5.

PROPOSITION 2 The characteristic function for the log price relative is

$$\begin{aligned} E \left(\exp \left(iu \ln \left(\frac{S(t)}{S(0)} \right) \right) \right) \\ = \exp \left(t \int_{-\infty}^{\infty} ((1 + y(x))^{iu} - 1) k(x) dx \right) \end{aligned} \quad (21)$$

Proof Consider the compensated jump martingale

$$n(u, t) = (\exp(iu \ln(1 + y(x))) - 1) * (\mu - \nu). \quad (22)$$

Let $N(u, t)$ be the stochastic exponential of $n(u, t)$. We have that

$$N(u, t) = \exp \left(\sum_{s \leq t} iu \ln(1 + \Delta Y(s)) \right)$$

$$\times \exp\left(-t \int_{-\infty}^{\infty} (e^{iu \ln(1+y(x))} - 1) k(x) dx\right) \quad (23)$$

It follows that

$$\begin{aligned} E\left(\exp\left(\sum_{s \leq t} iu \ln(1 + \Delta Y(s))\right)\right) \\ = \exp\left(t \int_{-\infty}^{\infty} (e^{iu \ln(1+y(x))} - 1) k(x) dx\right) \end{aligned} \quad (24)$$

The result follows on noting (17) and simplifying the right-hand side. ■

For the inverse logistic model, the unit time characteristic exponent $\psi_{IL}(u)$ is given by

$$\psi_{IL}(u) = \int_{-\infty}^{\infty} \left(\left(\frac{2}{1 + e^{-x}} \right)^{iu} - 1 \right) k(x) dx. \quad (25)$$

The required integral in the characteristic exponent may be numerically evaluated for a variety of models for the Lévy measure. We report on the bilateral gamma and the *CGMY* models.

The bilateral gamma model has four parameters for scale and speed of the process on the two sides. The scale parameters are b_p, b_n while the speed parameters are c_p, c_n with Lévy density

$$k_{BG}(x) = c_n \frac{\exp(-|x|/b_n)}{|x|} \mathbf{1}_{x < 0} + c_p \frac{\exp(-x/b_p)}{x} \mathbf{1}_{x > 0}. \quad (26)$$

The *CGMY* has two additional parameters y_p, y_n with $b_n = 1/G$ and $b_p = 1/M$ and Lévy density

$$k_{CGMY}(x) = c_n \frac{\exp(-|x|/b_n)}{|x|^{1+y_n}} \mathbf{1}_{x < 0} + c_p \frac{\exp(-x/b_p)}{x^{1+y_p}} \mathbf{1}_{x > 0}. \quad (27)$$

4. Results on daily returns

For the time series analysis, the bilateral gamma model was fit to data for x_t with returns y_t defined by

$$y_t = \frac{S_{t+1} - S_t}{S_t}$$

and x_t defined using equation (8) for the inverse logistic model and equation (10) for the inverse hyperbolic tangent model. The inverse exponential defined by

$$x_t = \ln(1 + y_t). \quad (28)$$

The object was to determine which transformation gave a better fit for the bilateral gamma model.

The data consisted of stock prices for 586 stocks for a thousand days ending 20201231. The estimation criterion minimized the difference between model and observed tail

Table 1. Quartiles of fit statistics.

	BG	BGIL1.0	BGIL0.5	BGatanh
Q1	0.02434	0.02460	0.02470	0.02464
Q2	0.03443	0.0352	0.03528	0.03534
Q3	0.04627	0.05084	0.05136	0.05191
Quartiles of RMSE				
Q1	0.00518	0.00521	0.00522	0.00521
Q2	0.00596	0.00605	0.00609	0.00607
Q3	0.00703	0.00733	0.00743	0.00741

Table 2. Parameter quartiles.

		b_p	c_p	b_n	c_n
BG	Q1	0.0073	1.3366	0.0103	1.0312
	Q2	0.0096	1.5790	0.0124	1.2315
	Q3	0.0121	1.8722	0.0152	1.4527
BGIL1.0	Q1	0.0155	1.2505	0.0204	1.0055
	Q2	0.0205	1.5543	0.0247	1.2244
	Q3	0.0277	1.8473	0.0307	1.4510
BGIL0.5	Q1	0.0310	1.2169	0.0412	0.9896
	Q2	0.0408	1.5495	0.0502	1.2166
	Q3	0.0552	1.8465	0.0621	1.4482
BGatanh	Q1	0.0071	1.2372	0.0105	1.0057
	Q2	0.0094	1.5510	0.0127	1.2258
	Q3	0.0120	1.8550	0.0158	1.4521

probabilities with Andersen and Darling (1952) weights. In addition, we evaluate the root mean square error for the tail probabilities.

Figures 1 and 2 present distribution functions of the two optimized fit statistics for the BG model estimated on the tail probabilities for four sets of data, the inverse logistic with bounds of unity and 0.5, inverse tanh and inverse exponential returns.

There is a significant improvement in the quality of the fit generated by the inverse logistic model with both bounds over both the inverse tanh and inverse exponential models. The return bound of 0.5 performed significantly better.

Instead of transforming the bounded daily simple return to a candidate increment of an underlying Lévy process with Lévy density of unbounded support and fitting the Lévy model to this transformed data one may employ the characteristic function of the logarithm of the stock to estimate the parameters of the process $X(t)$ consistent with the data on daily log returns. This second approach recognizes that it is instantaneous simple returns that are bounded, and not the daily simple return arising from the underlying continuous time model. The first approach treats daily returns as if they were instantaneous. Table 1 presents the quartiles of the fit statistic and the root mean square error for the four models applied to a thousand days of log daily returns on the 586 stocks ending 31 December 2022. The four models provide an adequate fit to data on daily log returns.

Table 2 presents quartiles for the parameter values of the underlying BG model for the four models.

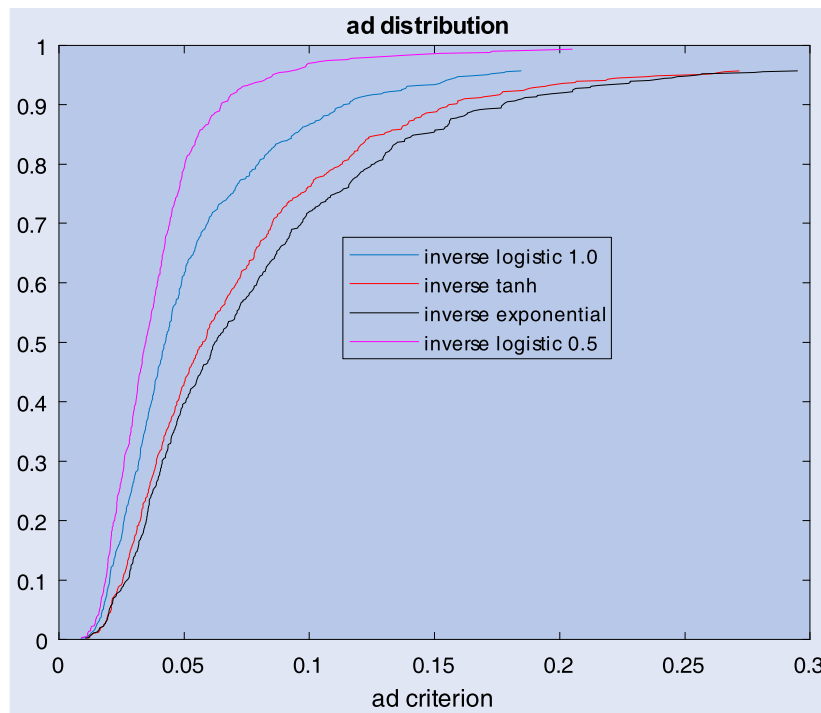


Figure 1. Distribution functions over 586 stocks for the bilateral gamma fitting criteria using Andersen Darling weights.

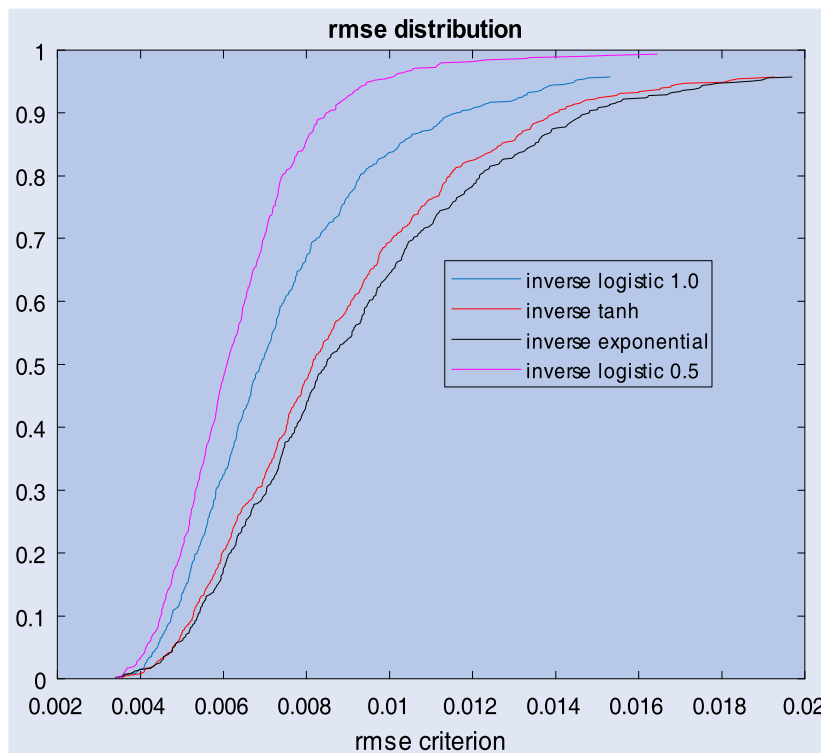


Figure 2. Distribution functions over 586 stocks for the bilateral gamma root mean square error.

5. Option pricing for revised returns

For return distributions embedded in option prices, the inverse logistic return model was fit every 10 days to option price data on *SPY* over the period January 2015 to December 2023 for a total of 238 estimation days. The model was fit separately to option price data at four exchange traded maturities closest to 1, 3, 6 and 12 months. The strikes employed with absolute

moneyness levels within 30 percent of at-the-money. Two versions of the inverse logistic model were employed, with return bounds at a 150 percent. Figure 3 presents the distribution of the average percentage error for the inverse exponential and two inverse logistic models at the four maturities.

A substantial improvement may be observed for the unit bound inverse logistic model that diminishes with the maturity. The inverse logistic with a return bound of 0.5 was also estimated but the performance was dominated by the unit

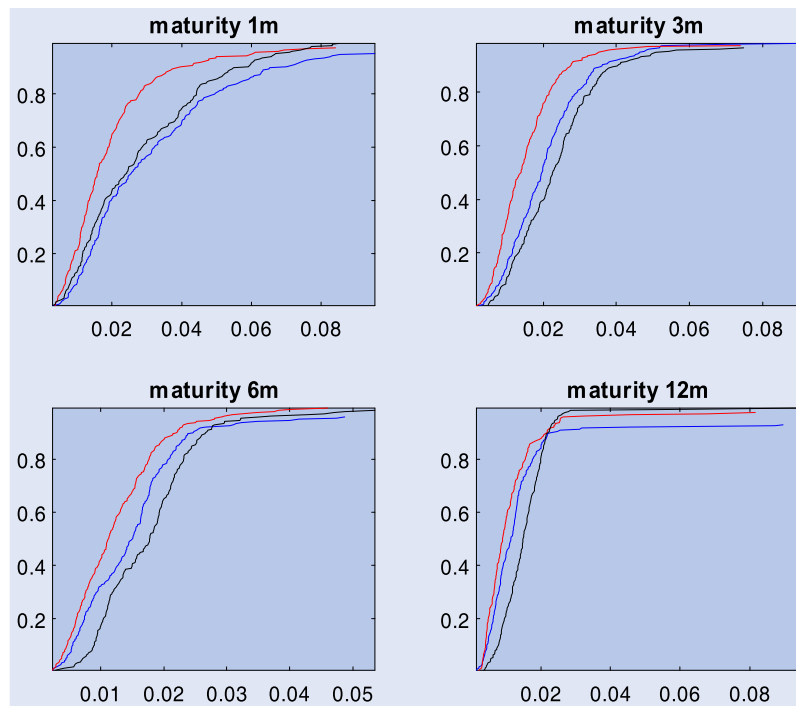


Figure 3. Distribution of the average percentage error for the inverse exponential in blue, the unit bound inverse logistic in red, and the 0.5 bound inverse logistic in black.

Table 3. No. of days with average percentage error below 3%.

Model	Maturity			
	1m	3m	6m	12m
BG	166	219	226	219
BGIL1.0	216	229	235	230
BGIL0.5	178	214	230	235

Table 4. Ratio of parameter standard deviation to the interquartile range.

Model	Parameter			
	b_p	c_p	b_n	c_n
BG	0.6934	113.17	0.6750	1.6506
BGIL1.0	1.4560	1.1527	0.7794	1.4081
BGIL0.5	6.6652	80.280	0.8014	1.9486

bound. With a view towards further understanding the model difference between the unity bound inverse logistic and the inverse exponential we evaluated the ratio of implied volatilities on each day at the 1 month maturity for a range of strikes with absolute moneyness below 15 percent. Figure 4 presents the three quartiles for these ratios at the various strike levels. The inverse exponential displays substantially higher implied volatilities for the upper quartile at the higher moneyness levels where the effect of the exponential is more pronounced.

5.1. Bilateral gamma model comparisons

For the three return definitions and the bilateral gamma model on the underlying shocks, table 3 reports for each maturity the number of days out of 238 on which the average percentage pricing error (APE) was below three percent. The inverse logistic model with unit bound generally has the greater number of days with average percentage errors below this threshold.

Restricting attention for each maturity for days on which the APE was below 3%, we evaluated the ratio of the standard deviation to the interquartile range. For a normal distribution this ratio is 0.74. It can reach high values when there are

Table 5. Parameter quartiles.

		b_p	c_p	b_n	c_n
BG	Q1	0.0037	7.9875	0.04503	5.4457
	Q2	0.0116	40.752	0.05624	8.8881
	Q3	0.0195	333.94	0.08674	13.558
BGIL1.0	Q1	0.0015	4.8200	0.08179	0.4454
	Q2	0.0031	242.36	0.09740	0.7294
	Q3	0.0174	641.32	0.13273	1.0643
BGIL0.5	Q1	0.0063	2.0316	0.1760	0.4706
	Q2	0.0267	6.2119	0.2174	0.6856
	Q3	0.0503	136.16	0.3035	1.0459

outliers or instabilities in the parameter estimations. Table 4 presents the ratios for the four parameters at the first maturity of one month where the instability is greatest. The ratios fall with the maturity.

By and large there is greater stability in the inverse logistic model with a unity bound on the return.

Table 5 presents quartiles of the four parameters for each of the return models.

The down side is well identified in all cases as is the upside scale. There is considerable variability in the upside speed with the unity bound inverse logistic decreasing scale and

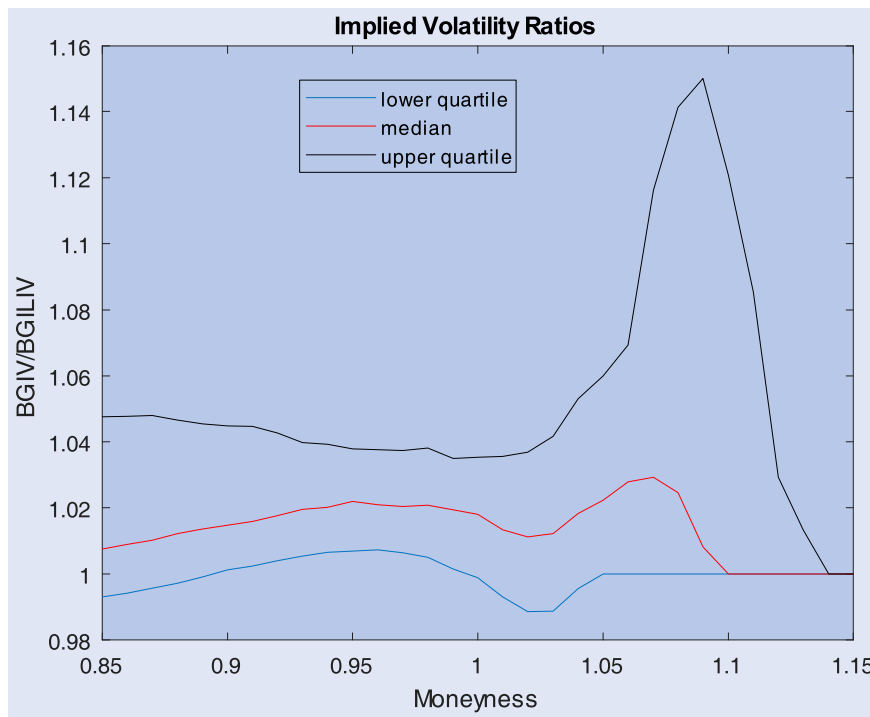


Figure 4. Quartiles for the ratio of inverse exponential implied volatility to the inverse logistic implied volatility at a range or strikes.

Table 6. Ratio of parameter standard deviation to the interquartile range for the cgmy model.

Model	Parameter					
	c_p	c_n	G	M	y_p	y_n
BG	1.2907	1.8217	0.8691	1.6186	0.5786	0.6353
BGIL1.0	1.7043	1.6890	0.8607	2.4613	0.5907	0.6164

Table 7. Parameter quartiles.

		c_p	c_n	G	M	y_p	y_n
BG	Q1	0.0096	0.0698	5.3020	10.5100	0.2566	0.6717
	Q2	0.0379	0.2513	7.0763	10.9558	0.5733	0.9889
	Q3	0.0904	1.0423	10.737	11.7144	1.2715	1.2272
BGIL1.0	Q1	0.0050	0.0152	3.3331	10.4726	0.2053	0.4388
	Q2	0.0138	0.0591	4.6335	10.9116	0.5488	0.8216
	Q3	0.0289	0.2866	7.2234	11.3172	1.1430	1.1935

lifting speed relative to the inverse exponential. On the other hand, the inverse logistic with a 50% return bound raises scale and lowers speed.

5.2. CGMY model comparisons

In addition to the bilateral gamma model, the *CGMY* model was estimated every 30 days for a total of 80 days on data for options on *SPY* at the four maturities separately. Two return definitions were used, the inverse exponential and the inverse logistic with a return bound of unity. Figure 5 presents the distribution function for the average percentage error for the two return definitions.

As may be observed the quality of fit is comparable with either return definition on the *CGMY* model. The number of days with an average percentage below 3% is comparable at each maturity and is not shown. Table 6 presents the ratio of the standard deviation to the interquartile range at the first maturity of one month.

Parameter quartiles are presented in table 7.

The stable alpha model of Mandelbrot (1963) was also estimated as the absence of exponential moments is not an issue for the inverse logistic. This is the *CGMY* model with

$G = M = 0$, $c_p = c_n = c$, and $y_p = y_n = y$. It is a two parameter model. However, the model was unable to fit option prices to any reasonable level for the average percentage error and the results are not reported. The minimum average percentage error was 13.73%. The restrictions imposed by the model are inconsistent with the option data.

6. Options on variations

This section develops the pricing of options on simple return absolute variations and quadratic variations. Let the simple return variation to time t be $V(t)$ while $Q(t)$ is the simple return quadratic variation. In our formulation, we have

$$V(t) = \sum_{s \leq t} |y(\Delta X_s)| \quad (29)$$

$$Q(t) = \sum_{s \leq t} (y(\Delta X_s)^2). \quad (30)$$

One may get to expectations and options on these entities via their distribution and hence via the associated characteristic functions.

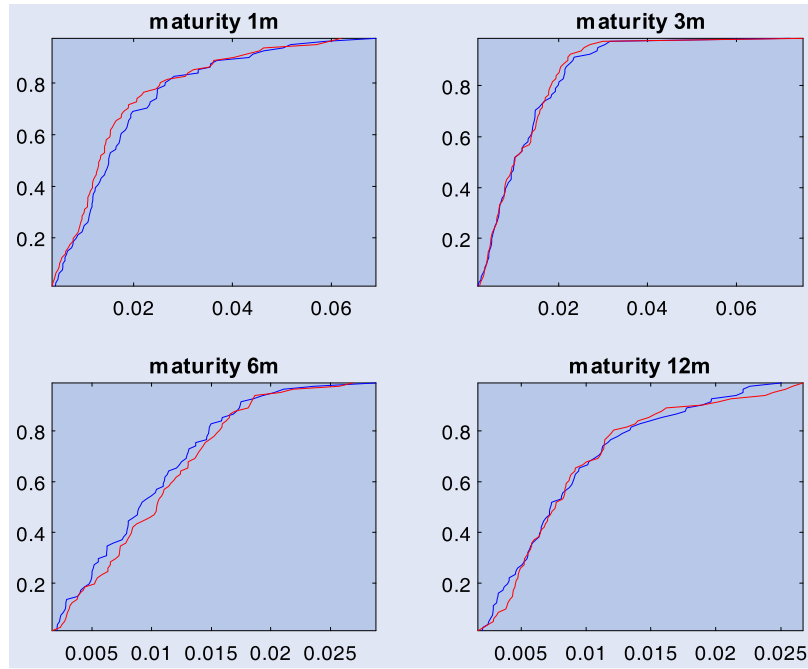


Figure 5. Average percentage error distribution functions for the fit of the inverse exponential and inverse logistic for the CGMY model. Exponential in blue and logistic in red.

Define the compensated jump martingales m_V, m_Q by

$$m_V(t) = (e^{iu|y(x)|} - 1) * (\mu - \nu) \quad (31)$$

$$m_Q(t) = (e^{iu(y(x))^2} - 1) * (\mu - \nu) \quad (32)$$

It follows on taking stochastic exponentials and expectations of m_V , and m_Q respectively that

$$\phi_V(u, t) = \exp \left(t \int_{-\infty}^{\infty} (e^{iu|y(x)|} - 1) k(x) dx \right) \quad (33)$$

$$\phi_Q(u, t) = \exp \left(t \int_{-\infty}^{\infty} (e^{iu(y(x))^2} - 1) k(x) dx \right) \quad (34)$$

In the inverse logistic case we have to evaluate the integrals

$$\begin{aligned} \psi_V(u, t) &= t \int_{-\infty}^{\infty} \left(e^{iu \left| \frac{2}{1+e^{-by}} - 1 \right|} - 1 \right) k(x) dx \\ &= t \int_0^{\infty} \left(e^{iu \left(\frac{2}{1+e^{-by}} - 1 \right)} - 1 \right) \frac{e^{-y}}{y} dy \\ &\quad + t \int_0^{\infty} \left(e^{iu \left(1 - \frac{2}{1+e^{by}} \right)} - 1 \right) \frac{e^{-y}}{y} dy \end{aligned} \quad (35)$$

Similarly for the quadratic variation, we evaluate

$$\begin{aligned} \psi_Q(u, t) &= t \int_0^{\infty} \left(e^{iu \left(\frac{2}{1+e^{-by}} - 1 \right)^2} - 1 \right) \frac{e^{-y}}{y} dy \\ &\quad + t \int_0^{\infty} \left(e^{iu \left(1 - \frac{2}{1+e^{by}} \right)^2} - 1 \right) \frac{e^{-y}}{y} dy \end{aligned} \quad (36)$$

These are well behaved and bounded integrands with integrations that may be accomplished by Trapezoidal methods once the risk neutral parameters for the process for $X(t)$ have been estimated. The results may be used to price options on simple return variations and quadratic variations.

6.1. Simple return absolute and variance options

We may determine

$$E[V(t)] = -i\phi'_V(0) \quad (37)$$

$$E[Q(t)] = -i\phi'_Q(0). \quad (38)$$

The price of the call option on $V(t)$ with strike a , $C(a)$ is

$$\begin{aligned} C(a) &= E[(V(t) - a)^+] \\ &= \int_a^{\infty} (v - a) f(v) dv \end{aligned} \quad (39)$$

Let $G(\lambda)$ be the Laplace transform of the call price on V in the strike level a .

$$\begin{aligned} G(\lambda) &= \int_0^{\infty} e^{-\lambda a} \int_a^{\infty} (v - a) f(v) dv da \\ &= \int_0^{\infty} dv f(v) \int_0^v (v - a) e^{-\lambda a} da \\ &= \int_0^{\infty} dv f(v) \left[v \left[-\frac{e^{-\lambda a}}{\lambda} \right]_0^v - \int_0^v a e^{-\lambda a} da \right] \\ &= \int_0^{\infty} dv f(v) \\ &\quad \times \left(\frac{(1 - e^{-\lambda v})v}{\lambda} - \left[-\frac{a e^{-\lambda a}}{\lambda} \right]_0^v + \int_0^v \frac{e^{-\lambda a}}{\lambda} da \right) \\ &= \int_0^{\infty} dv f(v) \left(\frac{(1 - e^{-\lambda v})v}{\lambda} + \frac{v e^{-\lambda v}}{\lambda} - \left[\frac{1 - e^{-\lambda v}}{\lambda^2} \right] \right) \\ &= \int_0^{\infty} dv f(v) \left(\frac{v}{\lambda} - \frac{1 - e^{-\lambda v}}{\lambda^2} \right) \\ &= \frac{E[v]}{\lambda} + \left(\frac{\Phi(\lambda) - 1}{\lambda^2} \right) \end{aligned}$$

$$= -\frac{\Phi'(0)}{\lambda} + \frac{\Phi(\lambda) - 1}{\lambda^2}, \quad (40)$$

where $\Phi(\lambda)$ is the Laplace transform of V . We may employ inverse Laplace transforms to evaluate call prices for simple absolute return and variance options.

It is best to work with a normalization to a unit expectation. Hence let $\theta = E[v]$ and set

$$E[e^{-\lambda v/\theta}] = \Phi\left(\frac{\lambda}{\theta}\right)$$

For the call option price transform of the normalized variate, we have

$$\tilde{G}(\lambda) = \frac{1}{\lambda} + \frac{\Phi(\lambda/\theta) - 1}{\lambda^2}$$

The strikes are then percentages of the expectation of unity. Similar expressions are valid for the call prices of options on quadratic variations.

7. Hedging considerations

In the classical case under sample path continuity, we have the replication result

$$Q = -2 \ln(S) + 2 \int_0^t \frac{dS}{S}. \quad (41)$$

We now have

$$\ln(S) = \int_0^t \ln(1 + y(x)) \mu(dx, ds) \quad (42)$$

$$V = \int_0^t |y(x)| \mu(dx, ds) \quad (43)$$

$$Q = \int_0^t y(x)^2 \mu(dx, ds). \quad (44)$$

Clearly these entities are not independent as they are constructed by different transformations of the same path space of the process for $X(t)$. To exploit the dependences that are present we proceed by simulation.

One may simulate N paths of length M for the bilateral gamma law for X into a matrix X that is $M \times N$. We then define the matrix Y by

$$Y = \frac{2}{1 + \exp(-X)} - 1. \quad (45)$$

We then generate the matrix $Z = \ln(S)$ as

$$Z = \ln(1 + Y)U, \quad (46)$$

where U is upper triangular with unit entries and Z is the matrix of cumulated sums of $\ln(1 + Y)$. The stock price paths are

$$S = \exp(Z + \omega), \quad (47)$$

Table 8. Call option prices.

Strike	AR	VAR
1	0.3122	0.6020
2	0.0992	0.4417
3	0.0323	0.3431
4	0.0103	0.2752

where ω is a convexity correction ensuring a unit exponential moment.

We also have

$$V = \mathbf{1}' \text{abs}(Y) \quad (48)$$

$$Q = \mathbf{1}' Y^2. \quad (49)$$

From the matrix S , we may form a matrix of returns over K subperiods with each period of length M/K . We denote this matrix by R . Let p be the vector of final stock prices in row M of S .

Let β be hedge positions in the subperiod returns that are to be determined. In addition to the dynamic subperiod hedge in analogy with the classic log contract hedge, we now seek a nonlinear function $\eta(p)$ as a static hedge. For the quadratic variation hedge, we perform a Gaussian Process Regression of the form

$$Q = R\beta + \eta(p) + \varepsilon, \quad (50)$$

The *GPR* to be estimated must employ a custom kernel that forces linearity in the variables R while using a nonlinear kernel for p . A similar hedge can be built for V the simple absolute return swap.

8. Variation options by simulation and Laplace inversion

For 29 March 2016, on *SPY* for the maturity of one month ($t = 0.0849$), the annualized bilateral gamma parameters were

$$b_p = 0.0245$$

$$c_p = 19.1280$$

$$b_n = 0.1260$$

$$c_n = 8.6148$$

Prices of call options for indicated strikes obtained by inversion of the Laplace Transforms for the normalized simple absolute return and its variance are presented in table 8.

8.1. Simulation and Laplace inversion results

On the path space for the logistic return being a bilateral gamma process, we compare the simple absolute and variance return with the variance swap paying the squares of the

Table 9. Call option implied volatilities.

Strike	simar		simvar		log. ret. var.
	simul	ILT	simul	ILT	
1	2.7512	2.7843	5.7476	5.5867	6.0589
2	2.5755	2.6150	5.5877	5.6941	5.9296
3	2.4494	2.4843	5.4763	5.5757	5.8373
4	2.3433	2.3694	5.3897	5.4824	5.7684

Table 10. Price reduction in vol. terms.

Multiple	simvar	log. ret. var
1	0.2241	0.2138
2	0.3354	0.3170
3	0.4142	0.3894
4	0.4753	0.4449

log return. The prices for the two variance contracts are, in volatility terms,

$$\text{simvar} = 19.58 \quad (51)$$

$$\text{var} = 20.37. \quad (52)$$

For multiples of 1, 2, 3, 4 of the variance as strikes, we report in table 9 the implied volatilities of call options on the simple return absolute return, the simple return variance, and the log return variance as determined on the inverse logistic of the simple return being a bilateral gamma process. For the simple return absolute and variance swaps, we present the simulated implied volatility and the implied volatility obtained by Laplace inversion. For the classical variance swap, the implied volatilities are by simulation.

Table 10 presents the percentage reduction in a fair variance swap price, quoted in volatility terms, for four multiples of the variance swap price, as capped variance swap strikes, for the simple return variance and the log return variance

9. Machine learning hedges

For the hedge of the simple absolute and squared return, the dynamic hedge took positions every 5 days for the 4 five periods to the 21 day maturity. For the variance of the log return with *GPR* nonlinear positioning, the dynamic hedge was also in four five-day returns. For the classical variance swap on the logarithmic return, the positioning was two dollars every day for 20 days. Figures 6–8 present the *GPR* hedge function in the first two cases and the log contract hedge in the last case. Also presented are the payouts net of the dynamic hedge.

Table 11 presents the dollar position in the four five period returns for *simvar* and the log return variance hedged by *GPR*.

The volatilities for dynamic hedges are as follows:

simvar	0.0016
vargpr	0.0058
varswap	0.1092

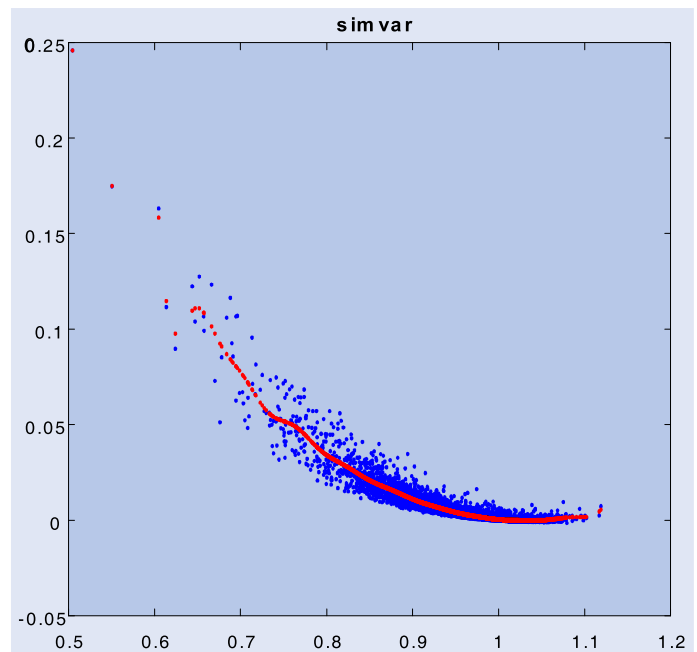


Figure 6. *GPR* hedge of the simple return variance swap and the associated payouts net of the four periodic dynamic hedge earnings.

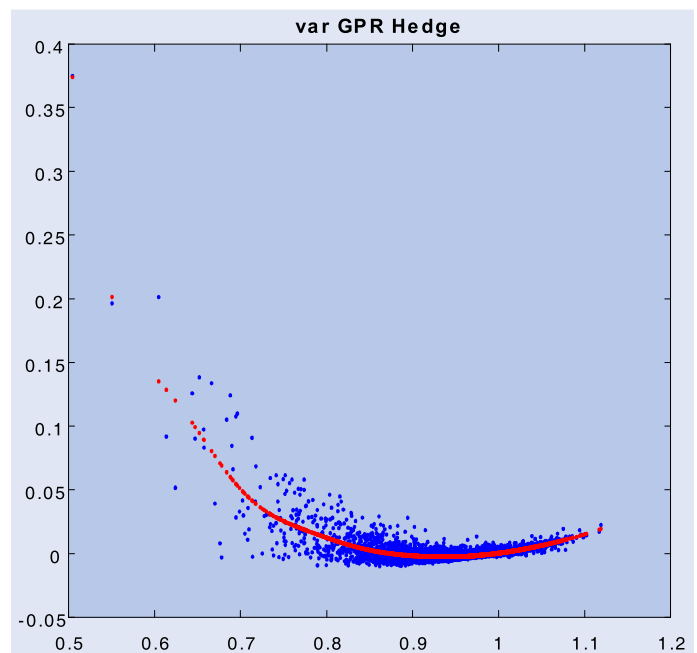


Figure 7. *GPR* hedge of the logarithmic return variance and payouts net of the four periodic dynamic hedge earnings.

It may be observed that the log contract dynamic hedge introduces a substantial volatility in the hope that the log contract can address this.

Figure 9 presents the hedge on the simple absolute return obtained by *GPR*.

9.1. Machine learning multiple maturities simultaneously

One may vectorize the cumulated squared variations at intermediate maturities and vectorize the stock prices at these

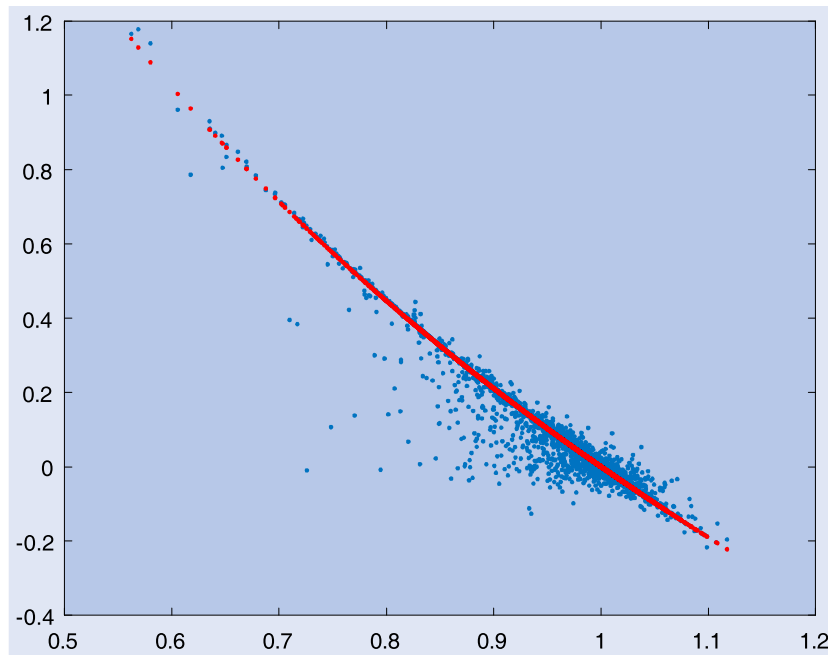


Figure 8. Log return variance hedged with the log contract.

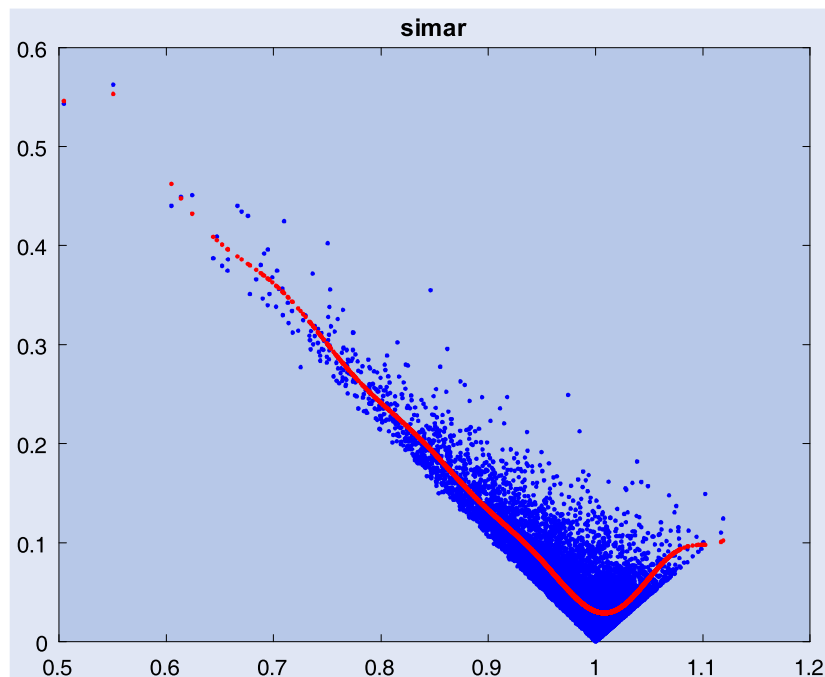


Figure 9. GPR hedge of simple absolute return.

Table 11. Dynamic dollar positions.

	Period			
	1	2	3	4
simvar	0.0305	0.03417	0.0262	0.0278
vargpr	-0.1042	-0.0970	-0.1129	-0.1098

maturities to perform a joint *GPR* at all four maturities seeking four hedge functions. We implemented the *GPR* hedge for four maturities simultaneously to find the same hedge function for all maturities. Different results could be obtained for more

complex processes that are not driven by processes of independent and identically distributed increments. This hedge functions are displayed in figure 10.

9.2. Time variation of ML hedging

With a view to assessing the variation over time of hedges based on Machine Learning, the 238 four bilateral gamma risk neutral parameters estimated for *SPY* options at a monthly maturity were first transformed to the levels for the mean and standard deviation of the upward and downward motions in the bilateral gamma process. The transformation delivers variables of a comparable order of magnitude suitable for

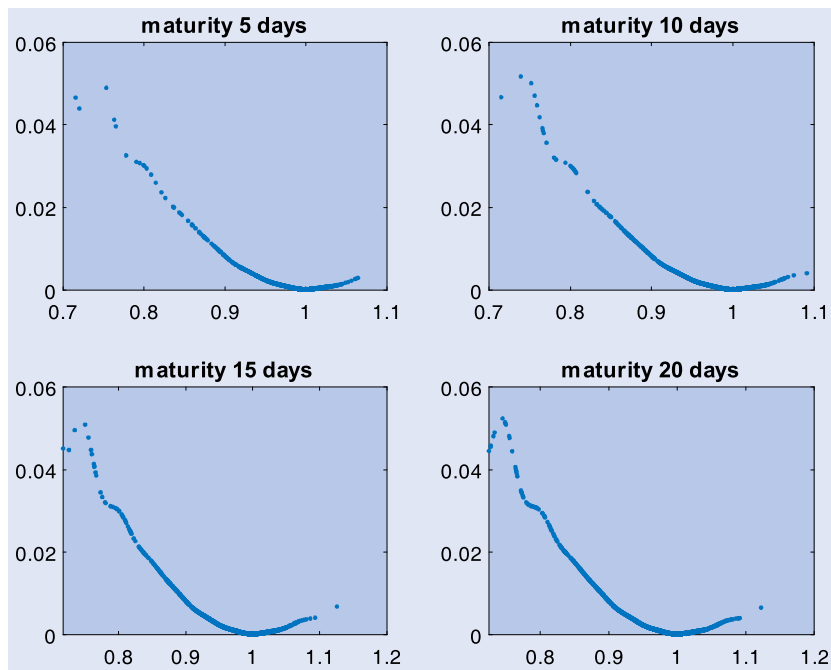


Figure 10. Simultaneous GPR hedging of four maturities for simvar on inverse logistic BG returns.

Table 12. Quantized BG parameters.

Case	Parameters				Proportion Represented
	b_p	c_p	b_n	c_n	
1	0.0018	794.07	0.1095	1.0043	7.77
2	0.0026	193.18	0.0986	0.6262	5.82
3	0.0022	358.25	0.0930	0.7722	19.42
4	0.0157	8.0405	0.1207	0.6853	10.68
5	0.0016	693.02	0.1061	0.7865	12.62
6	0.0065	50.158	0.1169	0.5030	5.83
7	0.0017	394.00	0.0881	0.6958	12.62
8	0.0152	3.6333	0.0908	0.8240	25.24

Table 13. Dynamic Hedges for simple return.

Case	Period			
	1	2	3	4
1	0.0234	0.0271	0.0333	0.0194
2	0.0050	-0.0002	-0.0023	-0.0040
3	0.0057	0.0122	0.0102	0.0125
4	0.0221	0.0279	0.0321	0.0304
5	0.0012	-0.0015	0.0017	-0.0004
6	0.0124	0.0014	-0.0023	0.0032
7	0.0005	0.0095	0.0074	0.0039
8	0.0105	0.0052	0.0069	0.0003

Table 14. Dynamic hedges for log return.

Case	Period			
	1	2	3	4
1	0.0008	0.0043	0.0112	-0.0044
2	-0.0565	-0.0490	-0.0439	-0.0410
3	-0.0109	-0.0017	-0.0034	-0.0014
4	-0.0754	-0.0607	-0.0553	-0.0651
5	-0.0067	-0.0099	-0.0047	-0.0053
6	-0.0364	-0.0524	-0.0551	-0.0519
7	-0.0140	-0.0009	-0.0065	-0.0097
8	-0.0388	-0.0450	-0.0441	-0.0530

quantization. They were quantized into a set eight representative vectors that were then transformed back into the original form for scale and speed of motion in the two directions. Table 12 presents the eight cases along with the proportion of points they represent.

Figure 11 presents the simple and log return variance hedges constructed by *GPR*. They differ markedly from the classical log contract hedge.

The dynamic hedge positions over the four five-day subperiods for the simple return variance and the log return variance for the eight cases are provided in tables 13 and 14.

Table 15 presents the volatilities across the simulated sample space for the annualized one month volatility of the dynamic hedge cash flow for the simple return variance, log return variance hedged by *GPR* and the log contract dynamic hedge.

Tables 16–18 present selected percentiles for the centered revenues from selling the simple return variance (*svar*), or the log return variance (*var*) hedged by dynamic trading and the static positioning in options as per the *GPR*. Also presented is the centered payout for the classical variance swap (*vswap*)

hedged by two dollars in the daily return and the log contract hedge. The three payouts are on a notional of 10 000 dollars.

10. Conclusion

Logarithmic returns are observed not to be consistent with absolute variations dominating quadratic variations, a

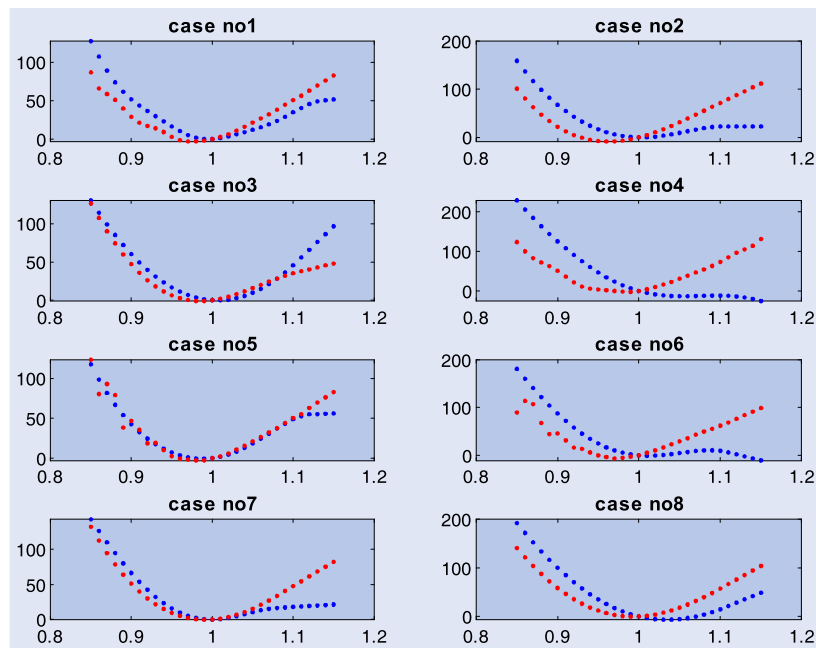


Figure 11. Eight cases of *GPR* constructed hedges for the one month simple return variance in blue and log return variance in red.

Table 15. Dynamic hedge volatilities.

Case	svar	var	log contract
1	0.0063	0.0015	0.4201
2	0.0005	0.0074	0.2924
3	0.0017	0.0010	0.3105
4	0.0055	0.0125	0.3844
5	0.0003	0.0014	0.3605
6	0.0011	0.0087	0.2807
7	0.0009	0.0014	0.2807
8	0.0010	0.0068	0.2976

Table 17. Revenue percentiles *var*.

Case	Percentiles				
	10	25	50	75	90
1	-22.0353	-5.0858	2.5030	9.0359	20.5066
2	-10.1642	-3.1672	0.8148	4.8621	10.3346
3	-11.6630	-2.8814	1.1731	5.1219	10.7017
4	-19.1628	-4.3934	2.2541	7.7152	17.5916
5	-14.4619	-3.4352	1.2699	5.6100	13.1640
6	-13.8504	-3.7208	1.3847	6.5973	13.3641
7	-7.9342	-2.1705	0.6376	3.7209	9.1897
8	-11.7750	-2.7366	1.4193	4.3329	10.3994

Table 16. Revenue percentiles *svar*.

Case	Percentiles				
	10	25	50	75	90
1	-23.0901	-5.1052	2.5314	8.7247	20.8720
2	-8.7129	-1.6767	0.8693	3.1698	8.9293
3	-11.4037	-3.0335	0.7768	4.6354	12.2299
4	-18.3364	-3.8528	2.7329	6.6255	14.6728
5	-14.1564	-2.8814	1.2177	4.7619	14.2067
6	-12.5509	-2.5834	1.7497	4.7736	12.0789
7	-8.0678	-1.5017	0.9659	2.8661	8.1455
8	-10.9431	-2.3434	1.5561	3.7545	9.1593

Table 18. Revenue percentiles *vswap*.

Case	Percentiles				
	10	25	50	75	90
1	-190.96	-114.47	-34.9883	48.2890	134.26
2	-134.07	-72.1309	-13.8970	38.9226	84.157
3	-152.53	-87.1257	-17.5339	47.0855	110.58
4	-167.22	-56.5165	4.0040	21.6713	28.5093
5	-152.56	-93.7288	-26.7977	37.0154	105.872
6	-158.72	-75.9123	-3.8927	49.8715	90.5296
7	-120.80	-67.7600	-14.6873	38.5757	85.5617
8	-87.026	-27.1875	-11.2125	-10.1621	16.9425

property empirically supported in the data. The domination is valid when returns measured by percentage changes are below unity in absolute value. Changes in real-valued processes are mapped to absolute returns bounded by unity using logistic and hyperbolic tangent transformations. Characteristic functions are developed for the log price relative when jumps in the underlying Lévy process effect stock prices with bounds on the percentage change in prices. The models are estimated on time series and option data and demonstrate improvements delivered by the inverse logistic return definition. Applications to pricing variations and options on them are developed.

Hedging strategies based on Machine Learning using Gaussian Process Regression (*GPR*) are inferred from simulated sample spaces. It is observed that the log contract hedge introduces a high volatility dynamic hedge that theoretically may be compensated by the log contract leaving the variance contract as a residual. The Machine Learned hedge works with other functions estimated here by *GPR* that substantially reduce the volatility of the dynamic hedge and yet leave as residual the variance payout.

Disclosure statement

No potential conflict of interest was reported by the author(s).

References

- Andersen, T.W. and Darling, D.A., Asymptotic theory of certain 'Goodness of fit' criteria based on stochastic processes. *Ann. Math. Stat.*, 1952, **23**(2), 193–212. <https://doi.org/10.1214/aoms/1177729437>.
- Bachelier, L., Théorie de la spéculation. *Ann. Sci. Ec. Norm. Supér. 3^e Série*, 1900, **17**, 21–86.
- Bachelier, L., *Le Jeu, la Chance et le Hasard*, 1914 (Ernest Flammarion: Éditeur, Paris).
- Barndorff-Nielsen, O.E., Processes of normal inverse Gaussian type. *Finance Stoch.*, 1998, **2**, 41–68.
- Black, F. and Scholes, M., The pricing of corporate liabilities. *J. Political Econ.*, 1973, **81**(3), 637–654.
- Broadie, M. and Jain, A., The effects of jumps and discrete sampling on volatility and variance swaps. *Int. J. Theor. Appl. Finance*, 2008, **11**(08), 761–797.
- Carr, P., Geman, H., Madan, D.B. and Yor, M., The fine structure of asset returns: an empirical investigation. *J. Bus.*, 2002, **75**(2), 305–332.
- Carr, P. and Madan, D.B., Towards a theory of volatility trading. In *Volatility*, edited by R. Jarrow, 1998 (Risk Publications: London).
- Carr, P. and Madan, D.B., Option valuation using the fast Fourier transform. *J. Comput. Finance*, 1999, **2**(4), 61–73.
- Carr, P. and Lee, R., Hedging variance options on continuous semimartingales. *Finance Stoch.*, 2010, **14**(2), 179–207.
- Carr, P., Lee, R. and Wu, L., Variance swaps on time-changed Lévy processes. *Finance Stoch.*, 2012, **16**(2), 335–355.
- Carr, P. and Wu, L., A tale of two indices. *J. Deriv.*, 2006, **13**(3), 13–29.
- Carr, P. and Wu, L., Variance risk premiums. *Rev. Financ. Stud.*, 2009, **22**(3), 1311–1341.
- Davis, M., Obloj, J. and Raval, V., Arbitrage bounds for weighted variance swap prices. *Math. Finance*, 2013, **24**(4), 821–854.
- Demeterfi, K., Derman, E., Kamal, M. and Zou, J., A guide to volatility and variance swaps. *J. Deriv.*, 1999, **6**(4), 9–32.
- Eberlein, E., Application of generalized hyperbolic Lévy motions to finance. In *Lévy Processes: Theory and Applications*, edited by O. E. Barndorff-Nielsen, T. Mikosch, S. Resnick, 2001 (Birkhäuser Verlag: Basel).
- Gatheral, J., *The Volatility Surface: A Practitioner's Guide*, 2006 (Wiley: New York).
- Küchler, U. and Tappe, S., Bilateral gamma distributions and processes in financial mathematics. *Stoch. Process. Their Appl.*, 2008, **118**, 261–283.
- Madan, D.B., Carr, P.P. and Chang, E., The variance gamma process and option pricing. *Rev. Financ.*, 1998, **2**(1), 79–105.
- Madan, D.B. and Seneta, E., The variance gamma (VG) model for share market returns. *J. Bus.*, 1990, **63**(4), 511–524.
- Mandelbrot, B., The variation of certain speculative prices. *J. Bus.*, 1963, **36**(4), 394–419.
- Merton, R.C., Theory of rational option pricing. *Bell J. Econ. Manage.*, 1973, **4**, 141–183.
- Neuberger, A., The log contract: A new instrument to hedge volatility. *J. Portf. Manag.*, 1994, **20**, 74–80.
- Samuelson, P.A., Rational theory of warrant pricing. *Ind. Manag. Rev.*, 1965, **6**, 13–31.